AUSTRONOMIC
Engineering the Cosmos

Austronomic Weekly Journal

Week 5: Focal Ratio and Exposure Efficiency

Field Notes | Photon Flux and f-ratio Geometry
2026-03-30 | v1.0
DOI: 10.5281/zenodo.19398649
ORCID: 0000-0003-4212-0230

Abstract

The focal ratio N, defined as focal length divided by aperture diameter and commonly written as f/N, is the primary determinant of focal-plane irradiance for spatially extended astronomical sources. For objects such as emission nebulae and galaxies, the focal-plane irradiance scales as source surface brightness divided by N squared, with total optical throughput acting as a multiplicative factor that cancels identically in any relative integration time comparison conducted at fixed camera and filter state. For unresolved point sources, aperture area rather than focal ratio governs the total collected signal; these two imaging regimes require distinct analytical frameworks, and conflating them produces systematic errors in exposure planning. The governing equations are instrument-independent and apply to any telescope-camera combination for which focal length and aperture diameter are known. A structured trade study applies all four governing equations to the Austronomic instrument collection; an experimental procedure on the Eta Carinae Nebula tests the predicted focal-plane irradiance ratio between the Askar 103APO native and reduced configurations by measuring diffuse nebular surface brightness in matched angular regions.

Keywords: *focal ratio; f-number; focal-plane irradiance; exposure efficiency; surface brightness; extended source; point source; relative integration time; optical throughput; etendue; astrophotography; telescope design*

Nomenclature

Symbol	Definition
ADU	Analog-to-Digital Unit
N	Focal ratio (dimensionless); N = f / D
f	Effective focal length (mm)
D	Aperture diameter (mm)
E	Focal-plane irradiance from an extended source (W m^{-2}) for a lossless optical system; multiplied by τ for real systems
L	Source surface brightness (W m^{-2} sr^{-1}); an intrinsic property of the source, independent of telescope aperture or distance
$\pi/4$	Geometric constant arising from the circular aperture; equals the projected aperture area $\pi*(D/2)^2$ divided by the focal-plane area $\pi*f^2$, giving the solid-angle factor $(\pi/4)/N^2$ per unit focal-plane area
τ	Total optical throughput of the imaging train (dimensionless); product of all transmission and reflection factors; cancels exactly in relative integration time comparisons at fixed camera and filter state
t_1, t_2	Integration times for configurations 1 and 2 under comparison (s)
N_1, N_2	Focal ratios of configurations 1 and 2 under comparison (dimensionless)
ROI	Region of Interest
s	Plate scale (arcseconds per pixel); s = 206.265 * p / f; for p in micrometers and f in millimeters.
sr	Steradian, the SI unit of solid angle; it measures angular area on the sky in three dimensions; sr is used in source surface brightness to indicate power received per unit collecting area per unit angular area on the sky.
p	Pixel pitch (micrometres)
206.265	Conversion constant: 1 radian = 206265 arcseconds; divided by 10^3 to reconcile pixel pitch in micrometres with focal length in millimetres; result: 206.265 arcsec mm um^{-1}

1. Introduction

Aperture determines how much light a telescope intercepts from a source, and focal length determines how that light is distributed across the focal plane as an image. The ratio of these two parameters, the focal ratio, governs the rate at which signal accumulates at each detector pixel from an extended celestial source. A proper treatment of focal ratio begins by separating two physically distinct imaging regimes: extended sources, whose surface brightness is an intrinsic property independent of distance, and point sources, whose signal on the detector depends primarily on aperture area.

For imaging extended objects such as emission nebulae and galaxies, the focal ratio is the dominant parameter controlling exposure efficiency. A smaller focal ratio concentrates the same collected light over a smaller projected area at the focal plane, delivering more photons per pixel per unit time. The governing equations developed here are instrument-independent and apply to any telescope-camera combination for which focal length and aperture diameter are known; any reader can substitute their own values throughout.

Sections 2 through 4 derive the four governing equations; Section 5 applies them in a trade study with an experimental validation plan on NGC 3372; Sections 6 and 7 address system trade-offs and connections to subsequent journals in this series.

2. The Focal Ratio: Definition and Physical Meaning

Consider a camera pointed at a uniformly lit overcast sky. If the lens is moved closer to the sensor, reducing the focal length while keeping the aperture fixed, each point on the sensor receives light from a wider patch of sky and therefore collects more light per second. As the focal length increases, the lens must sit further from the sensor, and the image of the same patch of sky is spread over a larger sensor area, so each point on the sensor receives proportionally less light per second. Crucially, the total light entering the camera does not change in either case: the lens still intercepts the same total beam. What changes is the angular area of sky that each point covers. The focal ratio captures this relationship in a single number: a smaller value means each sensor point covers more sky and receives more photons per unit time from any uniformly bright extended scene. The sections that follow replace this geometric intuition with exact equations.

2.1 Formal Definition

The focal ratio N is the ratio of focal length f to aperture diameter D, as stated in Equation (1). Both quantities carry the same unit, so N is dimensionless. The image formed at the focal plane is two-dimensional, so doubling the focal length doubles the linear image scale in both axes and therefore quadruples the focal-plane area over which the same total flux is distributed. This geometric dilution is the foundation of the 1/N squared relationship derived in Section 3 [1].

$$N = \frac{f}{D} \tag{1}$$

N = focal ratio (dimensionless)

f = focal length (mm)

D = aperture diameter (mm)

Equation (1) confirms that the focal ratio is fully determined by aperture diameter and focal length. Two instruments at the same f-ratio, regardless of physical size, produce the same focal-plane irradiance per unit detector area

from any extended source. The Askar 71F at f/6.9 and the Askar 91F APO at f/6.7 are therefore nearly matched in exposure speed for extended objects, while the Celestron EdgeHD 800 at f/10.0 is substantially slower for the same nebula target Equation (1) [1].

The definition captures the geometry of light concentration, but the quantitative link between focal ratio and pixel signal requires the focal-plane irradiance equation, which is derived in Section 3.

3. Focal-Plane Irradiance and the $1/N^2$ Law

The focal ratio governs exposure speed through its effect on focal-plane irradiance: the light power delivered per unit area at the detector surface. For extended sources this relationship takes a compact and exact form that underlies all exposure time analysis in astrophotography. For point sources it does not hold, and the two regimes require separate treatment. The derivation begins from the solid angle that the aperture subtends as seen from a point on the focal plane.

3.1 Extended Sources

For a source with surface brightness L, the focal-plane irradiance E is given by Equation (2). The π/4 coefficient arises from integrating the solid angle subtended by the circular pupil of diameter D at distance f: the result is $\pi*(D/2)^2/f^2 = (\pi/4)/N^2$, which appears as the denominator factor in the equation [2].

$$E = \frac{\pi}{4} \times \frac{L}{N^2} \qquad (2)$$

E = focal-plane irradiance for a lossless optical system (W m^{-2})

π/4 = geometric constant for a circular aperture; equals the projected aperture area $\pi*(D/2)^2$ divided by the focal-plane reference area $\pi*f^2$, giving the solid-angle factor $(\pi/4)/N^2$ per unit focal-plane area

L = source surface brightness (W m^{-2} sr^{-1})

N = focal ratio (dimensionless)

Equation (2) describes an idealised, lossless optical system. Real imaging trains attenuate the focal-plane irradiance by total optical throughput τ, the product of all transmission and reflection factors, giving the practical form $E_{practical} = (\pi/4)\tau L/N^2$. When comparing two configurations, τ cancels in the integration-time ratio only if the total optical throughput is equal between both configurations. Using the same camera and filter state guarantees that the detector quantum efficiency and filter transmission components of τ cancel exactly; the residual is the telescope optical train transmission, determined by the number of lens elements, mirror reflectivities, corrector plates, focal reducers, coatings, and central obstruction, which is instrument-specific and does not cancel across configurations of different optical design. The relative integration-time prediction is therefore strictly τ-independent only when the full optical train throughput is equal between configurations, as it is when comparing different focal-ratio settings of the same telescope. The $1/N^2$ dependence is the central result. Halving the focal ratio increases the focal-plane irradiance by a factor of four, while doubling it reduces the irradiance to one quarter, as shown by Equations (2) and (3) [2].

Three distinct signal quantities appear in the analysis and experiment, and they must not be conflated. First, focal-plane irradiance E (W m^{-2}) is the physical optical quantity that Equation (2) models. Second, the detected signal in photoelectrons per pixel per second equals E

multiplied by detector quantum efficiency and pixel area; this is the quantity that the camera integrates during an exposure. Third, the calibrated readout in analogue-to-digital units (ADU) per pixel equals the electron count divided by the sensor gain in electrons per ADU. For a fixed camera at fixed gain and filter state, all three quantities are proportional to one another; **ADU per pixel per second may therefore be used as a practical proxy for focal-plane irradiance in any relative comparison.**

Figure 1 illustrates the geometric basis of the $1/N^2$ law by showing how two different focal lengths, applied to the same aperture, concentrate or dilute the collected light over different focal-plane areas.

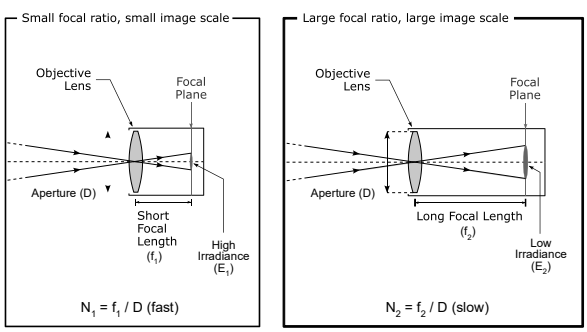

Figure 1. *The $1/N^2$ irradiance law illustrated geometrically. For a fixed aperture D, a shorter focal length (left) subtends a larger solid angle at the focal plane, producing higher irradiance E_1. A longer focal length (right) distributes the same collected light over a larger area, reducing irradiance to $E_2 = E_1 (N_1/N_2)^2$.*

Figure 1 makes clear that the aperture diameter is the same in both panels: the irradiance difference arises entirely from the change in focal length, not from any change in total collected light. The total photon flux intercepted by the telescope is proportional to aperture area and remains unchanged; what changes is how that flux is distributed across the detector.

3.2 Point Sources

For unresolved point sources such as stars, the $1/N^2$ irradiance law does not govern the signal in the same direct way. A star delivers its entire collected flux into a seeing-limited or diffraction-limited spot on the focal plane. The total signal accumulated in that spot per unit time is proportional to aperture area D^2: a larger aperture collects more photons independently of the focal ratio. The focal ratio determines how many pixels the point spread function spans, which affects signal per pixel and sky background noise per pixel, but it does not set the total signal integrated over the point spread function [3].

This distinction motivates different instrument selection criteria for different target classes. **High-resolution planetary imaging and fine stellar astrometry weight aperture above focal ratio.** Deep-sky nebula imaging, where every pixel accumulates signal proportional to $1/N^2$, weights a small focal ratio alongside aperture. Section 4 translates the irradiance result into the practical planning tool: the relative integration time.

4. Exposure Time Scaling

The $1/N^2$ irradiance law implies a direct relationship between focal ratio and the integration time required to reach a given signal level on an extended source. In Figure 1, if configuration 1 at focal ratio N_1 reaches a target signal level in integration time t_1, then configuration 2 at focal ratio N_2 requires integration time t_2 given by Equation (3), which follows directly from setting E_1 times t_1 equal to E_2 times t_2 and applying the proportionality E proportional to $1/N^2$. Because τ cancels in this ratio, Equation (3) holds for any two configurations sharing the same camera and

filter state, regardless of their absolute throughput values.

$$\frac{t_2}{t_1} = \left(\frac{N_2}{N_1}\right)^2 \quad (3)$$

t_1 = integration time for configuration 1 (s)

t_2 = integration time for configuration 2 (s)

N_1 = focal ratio of configuration 1 (dimensionless)

N_2 = focal ratio of configuration 2 (dimensionless)

Using the Askar 71F at f/6.9 as the reference (N_1 = 6.9), a configuration at f/10.0 requires $(10.0/6.9)^2$ = 2.10 times more integration time, while the Askar 103APO reduced to f/5.44 requires only $(5.44/6.9)^2$ = 0.62 times as much. More generally, a factor of sqrt(2) change in focal ratio corresponds to a factor of two in required integration time, which photographers recognise as one photographic stop Equation (3) [2].

Table 1 summarises the two imaging regimes and the mixed-field case by their governing parameters, confirming that the f-ratio squared rule applies strictly to extended objects and that mixed fields require the two components to be assessed separately.

Table 1. *Imaging regime classification by target type. The f-ratio squared rule in Equation (3) applies only to extended objects. Mixed fields require the diffuse and point-source components to be assessed independently.*

Target type	Governing parameter	Practical consequence
Extended object (nebula, galaxy, diffuse halo)	Focal ratio N; E = (π/4)*L/N^2	Faster f-ratio reduces relative integration time; aperture sets resolution and field coverage, not pixel brightness
Point source (unresolved star)	Aperture D; total flux proportional to D^2	Larger aperture increases signal; focal ratio determines PSF spread across pixels and sky background per pixel
Mixed field (globular cluster, galaxy with resolved stars)	Diffuse halo follows N; stellar peaks follow D	Separate the diffuse surface brightness and stellar components before comparing configurations; angular resampling to a common plate scale is required

Table 1 shows that focal ratio is a reliable exposure efficiency metric only when applied to the correct regime. For stellar photometry and astrometry, aperture area is the design parameter of interest, and focal ratio is chosen to achieve the required plate scale rather than to maximise pixel-level irradiance. Section 5 applies Equations (1) through (4) to the full instrument collection and presents the experimental method for directly measuring the irradiance ratio on a target extended source.

5. Instrument Trade Study

The three equations in Sections 2 through 4 organise any set of telescope-camera combinations into a coherent exposure efficiency hierarchy for extended-source imaging. The analytical component below applies this hierarchy to the Austronomic instrument collection; the experimental component provides the general measurement method and a specific worked example that any reader can adapt to their own equipment.

5.1 Analytical Component

The parameter under investigation is focal-plane irradiance for extended sources, governed by Equation (2). Equation (3) then converts the irradiance comparison into a practical relative integration time, allowing direct comparison of session time required by each configuration to reach a given signal threshold.

Table 2 also includes plate scale values for each configuration, providing the angular-sampling context that prevents focal ratio from being treated as the sole system descriptor. Equation

Table 2. *Focal ratio trade study for the Austronomic telescope-camera configurations (worked examples). All focal lengths and f-ratios are verified from instrument datasheets. Relative irradiance = (6.9/N)^2; relative integration time = (N/6.9)^2; both normalised to the Askar 71F + ASI585MC at f/6.9. Plate scale values verified from Equation (4) using datasheet pixel pitches. The methodology applies identically to any telescope-camera combination.*

Configuration (worked example)	FL (mm)	Aperture (mm)	Pixel (um)	f-ratio	Plate scale (arcsec/px)	Rel. irradiance	Rel. integration time
Askar 103APO + ASI533MC Pro (0.8x reduced)	560	103	3.76	f/5.44	1.38	1.61	0.62
Askar 91F APO + ASI533MC Pro	609	91	3.76	f/6.7	1.27	1.06	0.94
Askar 103APO + ASI533MC Pro (native)	700	103	3.76	f/6.8	1.11	1.03	0.97
Askar 71F + ASI585MC	490	71	2.90	f/6.9	1.22	1.00	1.00
EdgeHD 800 + ASI533MC Pro (0.7x reduced)	1422	203	3.76	f/7.0	0.55	0.97	1.03
EdgeHD 9.25 + ASI462MC (0.7x reduced)	1645	235	2.90	f/7.0	0.36	0.97	1.03
EdgeHD 800 + ASI533MC Pro (native)	2032	203	3.76	f/10.0	0.38	0.48	2.10
EdgeHD 9.25 + ASI462MC (native)	2350	235	2.90	f/10.0	0.25	0.48	2.10

(4) states the plate scale as a function of pixel pitch and focal length [1].

$$s = 206.265\frac{p}{f} \qquad (4)$$

s = plate scale (arcseconds per pixel)

206.265 = conversion constant: 1 radian = 206265 arcseconds; divided by 10^3 to reconcile pixel pitch in micrometres with focal length in millimetres

p = pixel pitch (micrometres)

f = effective focal length (mm)

Table 2 combines the results of Equations (1), (3), and (4), normalised to the Askar 71F + ASI585MC at f/6.9. The Askar 71F is chosen as the reference because it is the smallest-aperture instrument in the collection; all native configurations slower than f/6.9 therefore show relative integration time greater than 1.00, making the table directionally intuitive. Plate scale values are verified from Equation (4) using datasheet pixel pitches. Any reader may extend the table by substituting their own focal length and pixel pitch into Equations (1), (3) and (4).

Table 2 confirms that the two native EdgeHD configurations at f/10.0 require 2.10 times more integration time than the Askar 71F reference to accumulate the same diffuse nebular signal per unit detector area. That factor is independent of aperture: the EdgeHD 800 at 203 mm and the EdgeHD 9.25 at 235 mm differ substantially in clear diameter, but their focal-plane irradiance per pixel from an extended source is identical at the same focal ratio. For low-surface-brightness extended-source work under time-limited sessions, native f/10 geometry therefore requires a substantially larger time investment than the refractor configurations. It becomes the appropriate choice when fine angular scale is the primary requirement, or when the target is compact and high in surface brightness and the session length is not the binding constraint.

A second result in Table 2 is operationally important. The three refractor configurations spanning f/6.7 to f/6.9 cluster within 6 percent of one another in relative integration time, making exposure speed a weak discriminator among them. The decision is more properly governed by field of view, sampling, guiding tolerance, and target framing requirements. **Note that the reduced EdgeHD 800 at f/7.0 and plate scale 0.55 arcsec/px and the reduced EdgeHD 9.25 at f/7.0 and plate scale 0.36 arcsec/px share identical relative integration times, yet differ by a factor of 1.5 in angular scale; they are not**

interchangeable in session planning despite their equal efficiency ranking.

5.2 Experimental Component

The focal-plane irradiance ratio between two configurations can be verified experimentally by measuring the diffuse nebular surface brightness in a matched angular region of interest (ROI) in each configuration stack, after local sky subtraction and calibration. The sky background is the quantity to subtract, not the quantity to measure; the target quantity is the sky-subtracted nebular emission signal in calibrated ADU per pixel per second. For a fixed camera at fixed gain and filter state, this signal is directly proportional to focal-plane irradiance and its ratio between configurations should equal $(N_1/N_2)^2$ from Equation (3). The method applies to any two instruments for which the target is observable at comparable altitude and transparency during the session [4].

The primary worked example uses the Askar 103APO in its two available configurations: native f/6.8 with focal length 700 mm and plate scale 1.11 arcsec/px, and 0.8x-reduced f/5.44 with focal length 560 mm and plate scale 1.38 arcsec/px, both paired with the ZWO ASI533MC Pro (IMX533 sensor, pixel pitch 3.76 um). Equation (3) predicts a relative integration time of (6.8/5.44)^2 = 1.56; the native configuration requires 56 percent more integration time per sub-frame to accumulate the same diffuse nebular signal per pixel as the reduced configuration.

The primary observation target is the Eta Carinae Nebula (NGC 3372), centred at J2000 coordinates RA 10h 45m 08s, Dec -59 deg 52 min. NGC 3372 is one of the brightest extended emission nebulae accessible from southern Australia, providing a large region of approximately uniform hydrogen-alpha emission at angular scales well beyond both detector footprints. From a site latitude of -37.9 degrees south, it transits at 68 degrees altitude, ensuring a short air-mass path and minimal differential extinction between sub-sequences. The object is best positioned for observation between March and May, when it transits near local midnight.

General site selection criteria are a Bortle 4 or lower sky classification, a clear southern horizon to at least 30 degrees altitude, and a drive time within 60 to 90 minutes of metropolitan Melbourne, Victoria. Sites in the Macedon Ranges precinct, approximately 60 minutes northwest, offer Bortle 4 to 5 conditions with adequate southern horizon access. Sites in the Marysville and Toolangi area, approximately 75 to 90 minutes northeast, offer Bortle 3 to 4 conditions with improved darkness to the south and east. Readers in other regions should apply the same general criteria to identify a locally equivalent site.

1. Allow the telescope and camera to reach thermal equilibrium with the ambient air for a minimum of 45 minutes before beginning science frames.
2. Complete polar alignment and confirm residual error is below 1 arcminute before starting the science sequence.
3. Focus on a magnitude 2 to 4 reference star within 15 to 20 degrees of NGC 3372 and above 45 degrees altitude, using a Bahtinov mask. Confirm critical focus with a short FWHM measurement in the ASIAIR or equivalent acquisition software.
4. Slew to NGC 3372 and confirm the science field remains above 45 degrees altitude throughout the planned imaging window.
5. For configuration 1 (Askar 103APO native, f/6.8, FL = 700 mm with ZWO ASI533MC Pro): capture bias, dark, and flat calibration

frames. Then acquire a minimum of 20 light frames at 120 s per frame, targeting peak ADU at 30 to 40 percent of sensor full-well capacity per sub-frame.

6. Switch to configuration 2 (Askar 103APO + 0.8x reducer/flattener, f/5.44, FL = 560 mm with ZWO ASI533MC Pro). Verify reducer back-focus spacing with a calibrated spacer ring before inserting; incorrect spacing shifts the effective focal length and produces field illumination gradients not corrected by flat frames. Refocus from scratch using the Bahtinov mask, as the additional glass path of the reducer shifts the focus position by several millimetres. Check that field illumination is symmetric by comparing the median ADU in the four quadrants of a flat frame; asymmetry greater than 3 percent indicates tilt in the reducer seating and should be corrected before proceeding. Capture new flat frames specific to this configuration, then acquire 20 light frames at 120 s per frame.
7. In PixInsight, calibrate each sequence using the corresponding bias, dark, and flat masters via the ImageCalibration process. Apply SubframeSelector to reject sub-frames with FWHM or eccentricity more than two standard deviations from the sequence median. Integrate the accepted sub-frames using ImageIntegration with sigma-clipping rejection.
8. Register both integrated stacks to a common reference frame using PixInsight's StarAlignment process with the reduced-configuration stack as the reference. Then resample the native-configuration stack onto the plate scale of the reduced stack (1.38 arcsec/px) using the Resample process with Bicubic Spline interpolation and the plate scale ratio 1.38/1.11 = 1.243 as the scaling factor.
9. Define the ROI in the reduced-configuration stack (the reference frame after resampling). Select a rectangular region at least 5 x 5 arcminutes in angular extent, located at least 20 arcminutes from the bright Eta Carinae core, free of resolved stars, bright filaments, and intensity gradients exceeding 10 percent across the box. Map this angular ROI into the resampled native-configuration stack by applying the same world coordinate position; the physical pixel extent will differ by the resampling factor. Using PixInsight's Statistics process, measure the median ADU per pixel within the ROI for each configuration stack. Separately measure a local sky background region of equal angular size adjacent to the ROI but free of nebulosity. Subtract the sky background median from the ROI median to obtain the sky-subtracted nebular signal for each configuration. Divide each sky-subtracted signal by its respective total integration time (20 x 120 s = 2400 s) to obtain signal in ADU per pixel per second.
10. Compute the measured focal-plane irradiance ratio: R_meas = (sky-subtracted nebular signal at f/5.44 in ADU/px/s) divided by (sky-subtracted nebular signal at f/6.8 in ADU/px/s). Acceptance criterion: $1.41 < R_{meas} < 1.72$, corresponding to the predicted value of 1.56 within 10 percent tolerance.

A measured ratio within the acceptance band confirms the $1/N^2$ relationship for this configuration pair. Principal discrepancy sources are: differential vignetting if flat-field calibration is imperfect or the reducer is tilted; atmospheric transparency variation between the two sub-sequences; any ADU pedestal difference between calibration sessions; incorrect reducer back-focus spacing, which shifts the effective focal length and distorts the predicted ratio; and focus

drift between configurations if the ambient temperature changed significantly during the hardware swap. Repeat the flat-quadrant symmetry check if any of these factors is suspected [4].

Results derived from simultaneous side-by-side imaging with independent optical trains are strictly more reliable than sequential comparisons because they eliminate atmospheric transparency as a variable. Sequential comparisons remain acceptable when two conditions are both satisfied: the sky transparency is stable throughout the session (verified by checking that the full-width at half-maximum of unsaturated reference stars and the median background ADU in non-nebular regions show less than 5 percent variation between the two sub-sequences), and the session interval between configurations is less than 90 minutes so that atmospheric extinction change is negligible. If either condition is not met, the sequential result should be treated as indicative rather than confirmatory.

Figure 2 shows the sky field footprints of the two Askar 103APO configurations overlaid on an outline of NGC 3372, with the recommended ROI and sky background regions annotated.

Figure 2 also illustrates that the reduced configuration provides a wider field and that the ROI and sky background boxes are defined in the reduced-frame coordinate system, then mapped by angular position into the native-frame stack. Both footprints encompass the region of relatively uniform hydrogen-alpha emission southwest of the bright Eta Carinae core that is suitable for the nebular surface brightness ratio measurement.

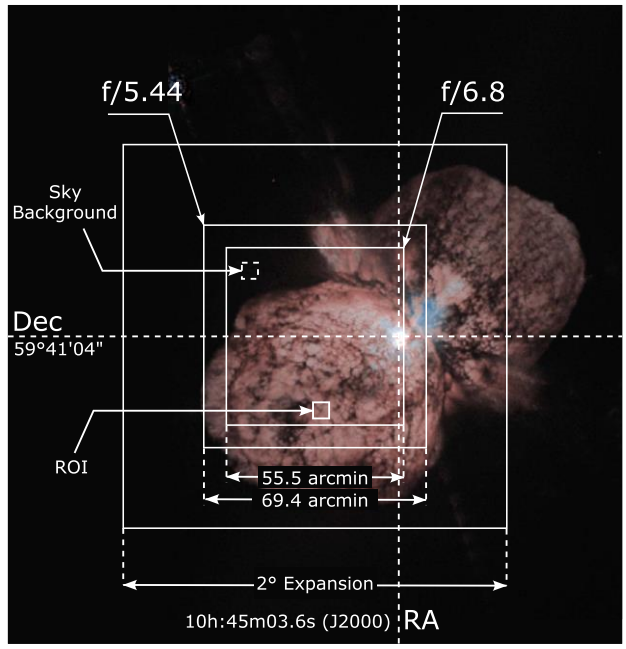

Figure 2. *Field footprints of the Askar 103APO at native f/6.8 (inner, 55.5 x 55.5 arcmin) and 0.8x-reduced f/5.44 (outer, 69.4 x 69.4 arcmin) overlaid on NGC 3372. The solid box marks the ROI for nebular surface brightness measurement; the dashed box is the adjacent sky background region. Both boxes are selected from the reduced-frame stack and mapped angularly into the native-frame stack [https://en.wikipedia.org/wiki/Eta_Carinae].*

6. System-Level Trade-Offs

The $1/N^2$ irradiance law creates a fundamental tension between two desirable imaging attributes: exposure speed and angular resolution. A small focal ratio delivers fast signal accumulation but produces a wide image at a coarse plate scale, as Equation (4) shows: for a fixed pixel pitch, plate scale coarsens as focal length decreases, and focal length decreases when the f-ratio decreases for a fixed aperture. Exposure speed and angular resolution move in opposite directions, and no single configuration satisfies both requirements simultaneously for all target types.

This trade-off is formalised by the concept of etendue, the product of the collecting aperture area and the field-of-view solid angle. Etendue is

a conserved quantity in a lossless optical system: any optical manipulation that increases the field of view proportionally reduces the solid angle of the collection pupil seen from the focal plane. The focal reducer on the Askar 103APO is a direct application of this principle: it simultaneously increases the field of view and raises the focal-plane irradiance per pixel by reducing N, but it cannot increase both the field of view and the angular resolution at the same time [5].

These considerations make the instrument collection coherent as an ensemble. The refractors at f/6.7 to f/6.9, with plate scales between 1.11 and 1.38 arcsec/px, are well matched to large extended nebulae that benefit from wide coverage and fast signal accumulation. The Schmidt-Cassegrain configurations at f/7.0 and f/10.0, with plate scales from 0.25 to 0.55 arcsec/px, are matched to compact objects requiring fine angular scale. Session planning begins by identifying whether the target's angular size and surface brightness favour a fast, wide-field configuration or a slower, high-resolution one, and selecting accordingly.

Sub-exposure length must also account for sky background: faster focal ratios fill the detector dynamic range more rapidly, constraining the maximum individual sub-exposure before sky glow dominates over read noise. The optimum sub-exposure length balances read noise against sky background and is configuration-specific. A complete exposure plan therefore combines the f-ratio squared rule for total integration time budgeting with a sky background noise analysis for individual sub-exposure length selection.

7. Forward Notes

The atmospheric seeing limit, examined in Week 6, interacts with focal ratio in a way not fully captured by the equations in this journal. A fast configuration with a coarse plate scale may concentrate the seeing-broadened star profile into fewer pixels, while a slower configuration at finer scale spreads the same profile across more pixels, affecting both peak signal and background noise per pixel in different ways. The Fried coherence length r_0 and atmospheric turbulence statistics introduced in Week 6 will quantify which plate scales can exploit the angular resolution that the optics could theoretically deliver.

The signal-to-noise framework in Week 9 will unify the focal-plane irradiance law, the aperture flux equation, and the detector noise model into a single expression for achievable signal-to-noise ratio as a function of integration time, aperture, focal ratio, and sky background. That framework will replace the proportionality arguments used in this journal with computable predictions, enabling direct comparison of achievable depth across all configurations for a specified target and sky condition.

References

[1] W. J. Smith, Modern Optical Engineering: The Design of Optical Systems, 4th ed. New York, NY, USA: McGraw-Hill, 2008.

[2] I. S. McLean, Electronic Imaging in Astronomy: Detectors and Instrumentation, 2nd ed. Berlin, Germany: Springer-Praxis, 2008.

[3] D. J. Schroeder, Astronomical Optics, 2nd ed. San Diego, CA, USA: Academic Press, 2000.

[4] R. Berry and J. Burnell, The Handbook of Astronomical Image Processing, 2nd ed. Richmond, VA, USA: Willmann-Bell, 2005.

[5] C. R. Kitchin, Astrophysical Techniques, 6th ed. Boca Raton, FL, USA: CRC Press, 2013.

AUSTRONOMIC

Engineering the Cosmos

Author Biography

Farzad Farajizadeh (Farzad Faradjizadeh) is an electrical engineer with a doctoral background and a multidisciplinary technical foundation spanning power systems, power electronics, electromagnetic compatibility, and related engineering domains. Through Austronomic, he explores astronomy from a systems-oriented perspective, with a particular emphasis on connecting physical principles, instrumentation, and engineering reasoning in a form accessible to motivated learners.

His broader objective with Austronomic is to develop a structured pathway into astronomy that moves beyond equipment-driven decision making and instead begins with conceptual clarity, technical rigor, and long-term system thinking. This work reflects his interest in building an educational framework in which astronomy can be approached not only as an observational pursuit, but also as a disciplined engineering and scientific practice.

© 2026 Farzad Faradjizadeh, Austronomic. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).
https://creativecommons.org/licenses/by/4.0/